DAY 03 • STUDENT HANDOUT

The Cyber Threat Landscape and Attack Frameworks

Malicious software behaviors, threat actors, the Cyber Kill Chain and MITRE ATT&CK

Qualification	Cyber Threat Monitoring Level I
Unit of Competency	CS-ICT251101 Monitor and Report Cyber Threats
Coverage	Required Knowledge 1.10 / 5.1 Malicious software behaviors; 5.2 Threats; 1.11 / 5.3 Knowledge of attack frameworks
Duration	8 hours (1 training day)

Learning Objectives

At the end of this session, you should be able to:

1. Identify common types of malware and describe their typical behaviors.
2. Describe the main categories of threat actors and their motives.
3. Explain the seven phases of the Cyber Kill Chain.
4. Navigate MITRE ATT&CK tactics and map observed behavior to a tactic and technique.
5. Recognize indicators of compromise (IOCs) and indicators of attack (IOAs).

1. Malicious Software (Malware) and Its Behaviors

Malware is software designed to harm, steal or gain unauthorized control. As a monitor, you rarely see the malware itself — you see its **behavior** in alerts and logs. Learning the behaviors helps you judge how serious an alert is.

Type	What It Does	Behaviors You May See in Alerts/Logs
Virus	Attaches to files and spreads when they are run	Many modified executables; AV detections on shared files
Worm	Spreads by itself across the network	One host connecting to many hosts on SMB (445)/RDP; sudden traffic spike
Trojan	Pretends to be legitimate software	User runs "invoice.exe"; new unknown process starts at boot
Ransomware	Encrypts files and demands payment	Mass file renames (e.g., .locked), shadow copy deletion (`vssadmin delete shadows`), ransom note files
Spyware / Infostealer	Steals passwords, cookies, keystrokes	Access to browser credential files; uploads to unknown sites
Backdoor / RAT	Gives the attacker remote control	Regular outbound "beacon" connections at fixed intervals
Botnet agent	Makes the host part of an attacker-controlled network	Connections to command-and-control (C2) servers; spam or DDoS traffic
Cryptominer	Uses the host CPU to mine	Constant high CPU; connections to mining pools

Type	What It Does	Behaviors You May See in Alerts/Logs
	cryptocurrency	
Fileless malware	Runs in memory using built-in tools (PowerShell, WMI)	Encoded PowerShell commands; Office app spawning a script engine
Rootkit	Hides itself and other malware deep in the OS	Security tool disabled or tampered; drivers loaded unexpectedly

Common Behaviors (What Malware Does After Landing)

- **Persistence** — survives reboot via Run registry keys, scheduled tasks, services or startup folders.
- **Privilege escalation** — tries to become Administrator/root.
- **Defense evasion** — disables antivirus, clears logs, uses encoded commands.
- **Credential access** — dumps passwords from memory (e.g., LSASS).
- **Discovery** — runs `whoami`, `net user`, `ipconfig` to learn about the environment.
- **Lateral movement** — moves to other hosts using stolen credentials, SMB, RDP or PsExec.
- **Command and control (C2)** — contacts the attacker's server for instructions.
- **Exfiltration / Impact** — steals data or encrypts/destroys it.

2. Threat Actors

Actor	Motive	Example
Cybercriminals	Money	Ransomware gangs, e-wallet phishing syndicates
Nation-state / APT groups	Espionage, disruption	Long-term intrusions into government agencies
Hacktivists	Political or social message	Website defacements, DDoS on government sites
Insiders	Revenge, money, carelessness	Employee leaking customer data
Script kiddies	Fun, reputation	Running downloaded tools against random sites

Common Initial Access Methods

- Phishing emails with malicious links or attachments
- Exploiting unpatched internet-facing systems (VPN, web servers)
- Stolen or weak passwords, especially on RDP/VPN
- Malicious USB drives and pirated software
- Compromised suppliers or third-party software

3. The Cyber Kill Chain

The **Cyber Kill Chain** (Lockheed Martin) describes an attack as seven phases. Detecting and stopping an attack at an early phase reduces damage.

#	Phase	Attacker Activity	What Defenders May Detect
1	Reconnaissance	Researches targets, scans for open ports	Port scans, web vulnerability scanner hits
2	Weaponization	Builds malware + exploit (e.g., macro	Usually not visible to the victim

#	Phase	Attacker Activity	What Defenders May Detect
		document)	
3	Delivery	Sends phishing email, malicious link, USB	Email gateway alerts, blocked attachments
4	Exploitation	Triggers the vulnerability or tricks the user	Office spawning PowerShell, exploit alerts
5	Installation	Installs malware and persistence	New services, scheduled tasks, Run keys
6	Command & Control	Malware contacts attacker server	Beaconing, connections to known-bad IPs/domains
7	Actions on Objectives	Steals, encrypts, destroys data	Large outbound transfers, mass file encryption

KEY POINT

The Competency Standards require you to report threats based on **exploitation activity and installation behavior** (phases 4–5), **spread and lateral movement**, and **egress and ingress** traffic. The Kill Chain helps you describe where the attack is.

4. MITRE ATT&CK Framework

MITRE ATT&CK is a free, global knowledge base of attacker **tactics** (the "why" — the goal) and **techniques** (the "how"). Each technique has an ID such as **T1059.001** (Command and Scripting Interpreter: PowerShell). SOC tools and reports use ATT&CK IDs as a common language.

Tactic (Enterprise)	Goal	Example Technique
Reconnaissance	Gather information	T1595 Active Scanning
Resource Development	Prepare infrastructure	T1583 Acquire Infrastructure
Initial Access	Get in	T1566 Phishing
Execution	Run code	T1059 Command and Scripting Interpreter
Persistence	Stay in	T1547 Boot or Logon Autostart Execution
Privilege Escalation	Get higher rights	T1068 Exploitation for Privilege Escalation
Defense Evasion	Avoid detection	T1562 Impair Defenses
Credential Access	Steal credentials	T1003 OS Credential Dumping
Discovery	Learn the environment	T1087 Account Discovery
Lateral Movement	Move to other systems	T1021 Remote Services
Collection	Gather target data	T1005 Data from Local System
Command and Control	Communicate with victim	T1071 Application Layer Protocol
Exfiltration	Steal data	T1041 Exfiltration Over C2 Channel
Impact	Disrupt / destroy	T1486 Data Encrypted for Impact

5. Indicators

- **IOC (Indicator of Compromise)** — evidence that something bad already happened: a malicious file hash, a known C2 IP or domain, a ransom note file.
- **IOA (Indicator of Attack)** — behavior showing an attack in progress: Word launching PowerShell, many failed logins followed by success, shadow copy deletion.
- IOCs change easily (attackers change IPs); IOAs (behaviors) are harder for attackers to change.

Key Terms

Term	Meaning
Malware	Software designed to harm, steal or gain unauthorized control
C2	Command and control — attacker infrastructure that sends instructions to malware
Persistence	Technique that lets malware survive a reboot
Lateral movement	Moving from one compromised host to others in the network
Tactic / Technique	ATT&CK terms: the attacker's goal / the method used to achieve it
IOC / IOA	Evidence of compromise / behavior of an attack in progress

References

- TESDA Competency Standards — Cyber Threat Monitoring Level I (ICT Sector), units CS-ICT251101 and CS-ICT251102.
- NIST SP 800-115, Technical Guide to Information Security Testing and Assessment (Chapter 4).
- NIST SP 800-61 Rev. 2/Rev. 3, Computer Security Incident Handling Guide.
- MITRE ATT&CK® Framework — attack.mitre.org; Lockheed Martin Cyber Kill Chain® .
- Republic Act No. 10175 (Cybercrime Prevention Act of 2012) and Republic Act No. 10173 (Data Privacy Act of 2012).